

homework-2

Packages

Question 1

```
install.packages("aRtsy")
```

The difference between `install.packages` and `library` is that `install.packages` downloads a package to the computer while `library` loads an already-installed package into the current session.

Question 2

```
library("aRtsy")
```

The Collatz Conjecture is a famous unsolved math problem that asks whether repeating two simple rules will always reach the number 1 for any positive whole number.

If the number is even, divide it by 2 until you get 1. For example, $4/2 = 2/2 = 1$ If the number is odd, multiply by 3, add by 1, and use the previous property until you get 1. For example, $1 * 3 + 1 = 4/2 = 2/2 = 1$

Question 3

Part A

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



This `canvas_collatz` function creates 10 blue lines on a blank “canvas” on the computer. These lines probably involve those Collatz Equations. Maybe it traces the path of 10 numbers using the Collatz equations.

Part B

- 1) You get a different output everytime you run the code because the function randomly chooses 10 numbers for the starting points.
- 2) This is difficult because everytime I render the document, the artwork will change. This potentially messes up my description of the artwork.

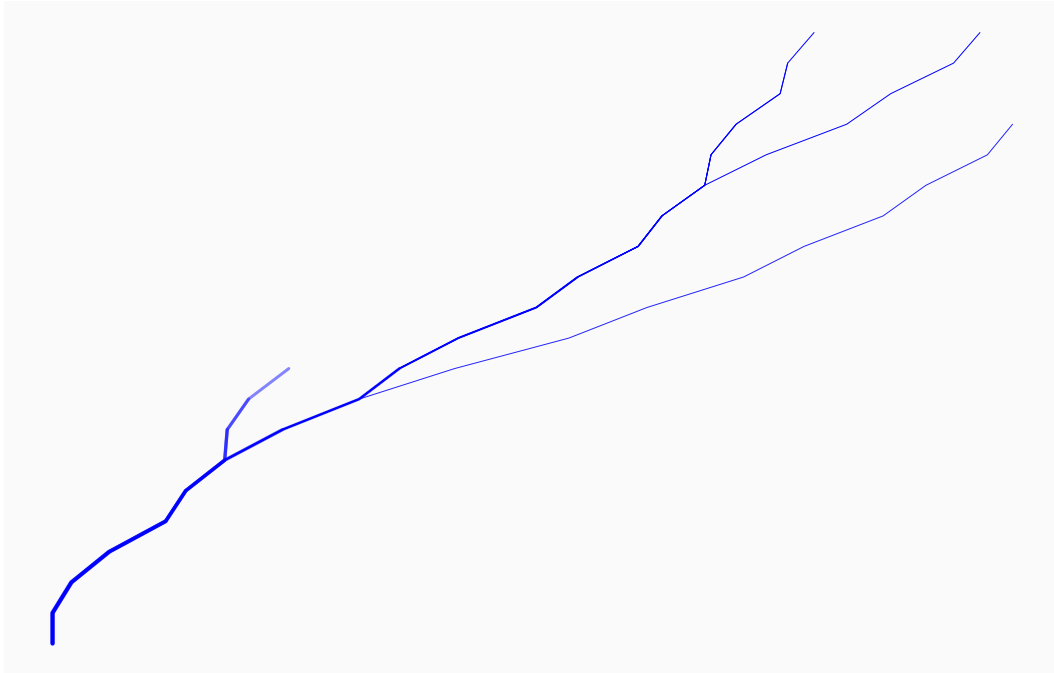
```
?set.seed # By giving it a fixed number, you force the random number generator to start at the  
set.seed(67)
```

Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



The first code randomly chooses any 10 integers while the second code uses the numbers 1 through 20. The second code always produces the same artwork while the first code's artwork changes after every "run" due to the random number generator.

Because $n = 1:20$ explicitly tells the code which numbers to use, the `set.seed` function is unnecessary.